Inverse relations and functions

1 $y = x$

2

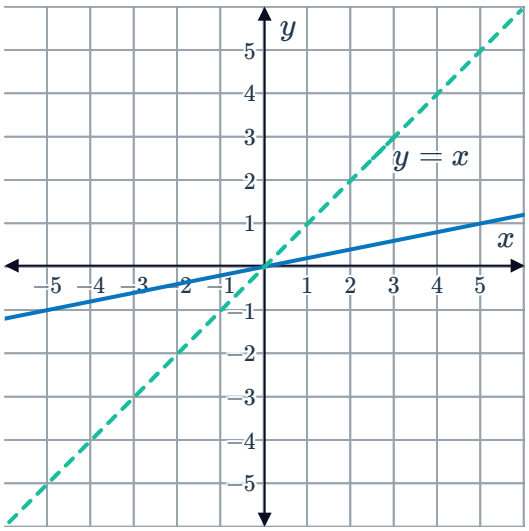
a No	**b** Yes	**c** No	**d** Yes
e No	**f** Yes	**g** No	**h** Yes

3

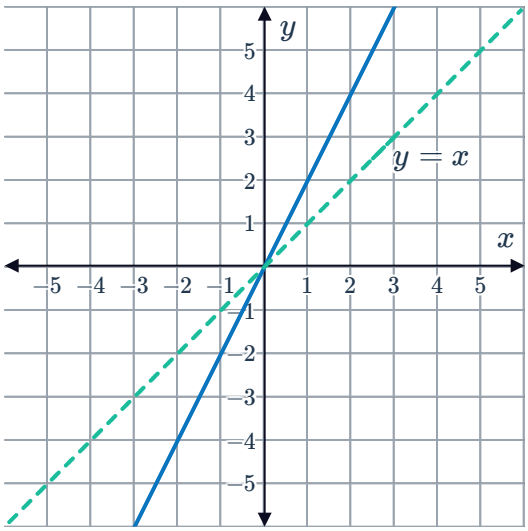
a Yes	**b** No	**c** Yes	**d** Yes
e No	**f** No	**g** Yes	**h** Yes
i Yes	**j** Yes	**k** No	

4

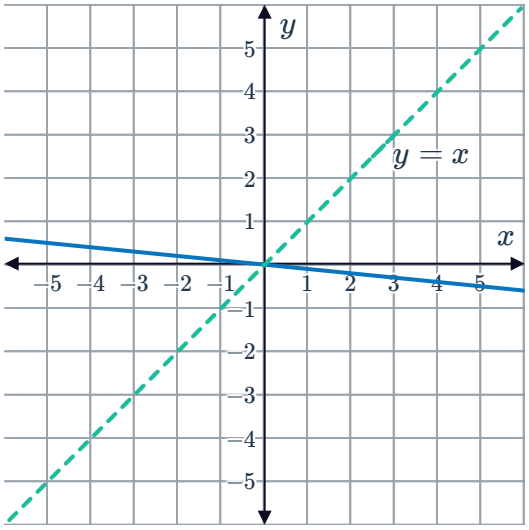
a



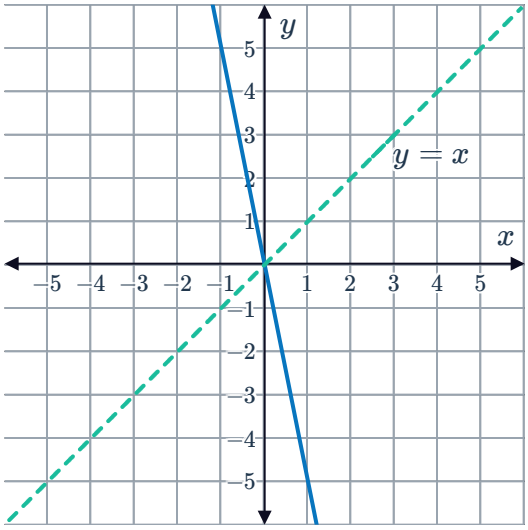
b



c

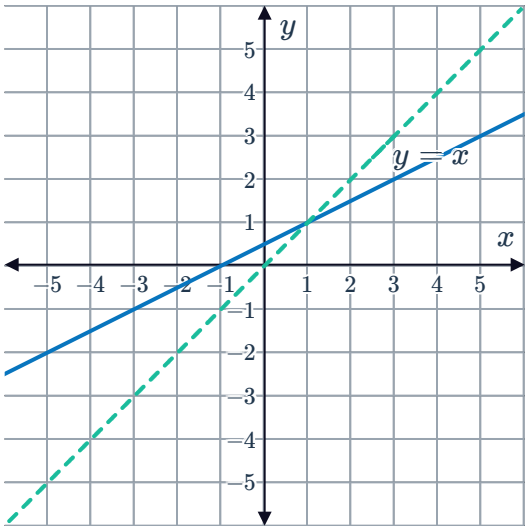


d

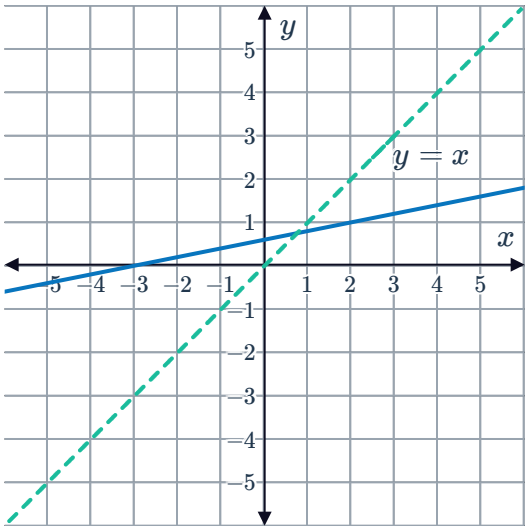


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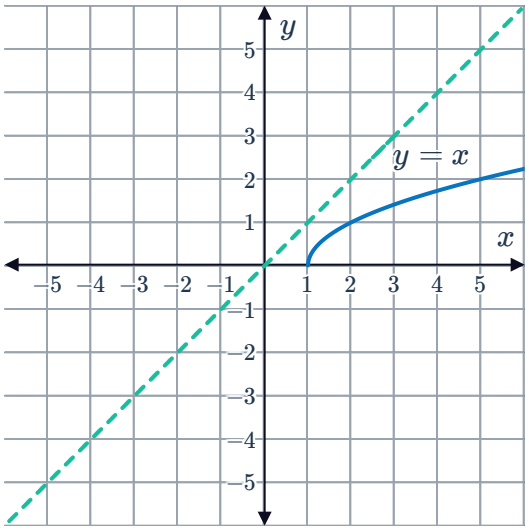
e



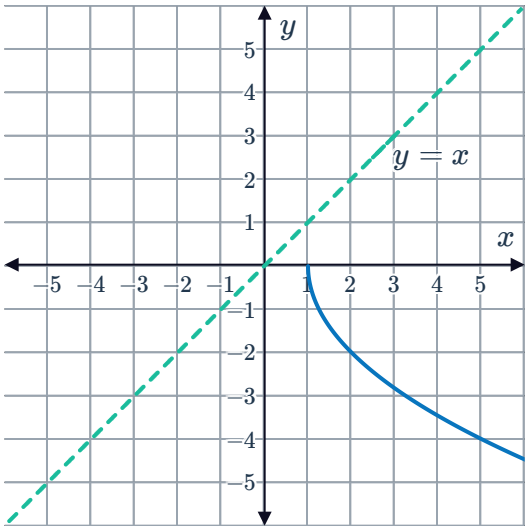
f



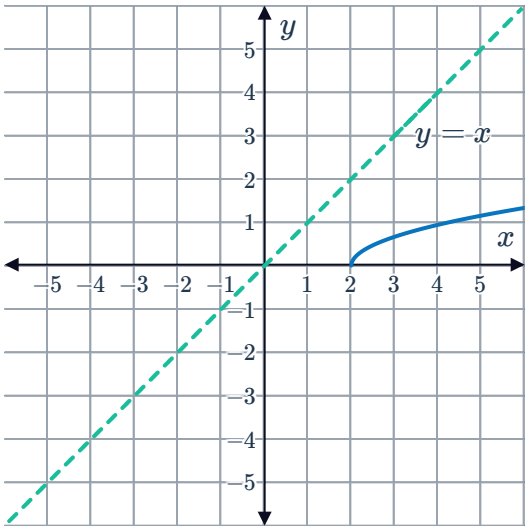
g



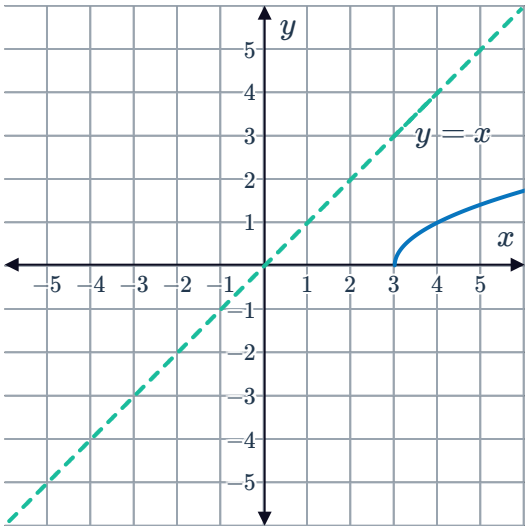
h



i



j



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5

a Function

b Relation

6

a

i x **ii** x **iii** Yes

b

i $x - 12$ **ii** $x - 4$ **iii** No

c

i x **ii** x **iii** Yes

7

a $y = \frac{x}{5}$

b $y = \frac{x + 8}{7}$

c $y = \frac{3}{4}(x + 5)$

d $y = \sqrt[3]{\frac{x}{6}}$

e $y = 8(1 - x)$

f $y = 6x + 7$

g $y = x^3 + 2$

h $y = \sqrt[3]{x - 3} + 4$

i $y = -\frac{2x}{x - 5}$

j $y = \frac{8}{x - 5} + 9$

k $y = \frac{8}{5x} - \frac{3}{5}$

l $y = \sqrt[3]{\frac{7}{x + 3}} + 8$

m $y = \frac{\sqrt[3]{\frac{x + 7}{4}} + 2}{9}$

Domain and range

8

a $\{-3\}$

b $\{6\}$

c $(6, -3)$

d $\{6\}$

e $\{-3\}$

f The domain of f is equal to the range of f^{-1}, and the range of f is equal to the domain of f^{-1}.

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9

a $\{6, -5\}$ **b** $\{6, -5\}$

10

a $\{5, 7, 8\}$ **b** $\{9, -4, 2\}$

c $\{(9, 5), (-4, 7), (2, 8)\}$ **d** $\{9, -4, 2\}$

e $\{5, 7, 8\}$

11

a $a = -\frac{1}{2}, b = \frac{5}{2}$ **b** $[5, 0]$ **c** $[-5, 5]$

12

a

i $f^{-1}(x) = \frac{1}{5}(x - 15)$ **ii** $(-\infty, \infty)$

iii $(-\infty, \infty)$

b

i $f^{-1}(x) = \frac{1}{3}(4 - x)$ **ii** $(-\infty, \infty)$

iii $(-\infty, \infty)$

c

i $f^{-1}(x) = \frac{1}{8}(x + 9)$ **ii** $[-41, 7]$

iii $[-4, 2]$

d

i $f^{-1}(x) = \sqrt{x}$ **ii** $[0, \infty)$

iii $[0, \infty)$

e

i $f^{-1}(x) = \sqrt{16 - x^2}$ **ii** $[0, 4]$

iii $[0, 4]$

13

a

i $f(x) = \frac{2x}{3} - 2$ **ii** $g(x) = \frac{3x}{2} + 3$

b

i x **ii** x

c

i True **ii** True **iii** False **iv** True

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14

a $[-4, 9]$ **b** $[-4, 5]$ **c** $[-4, 5]$ **d** $[-4, 9]$

e False

15

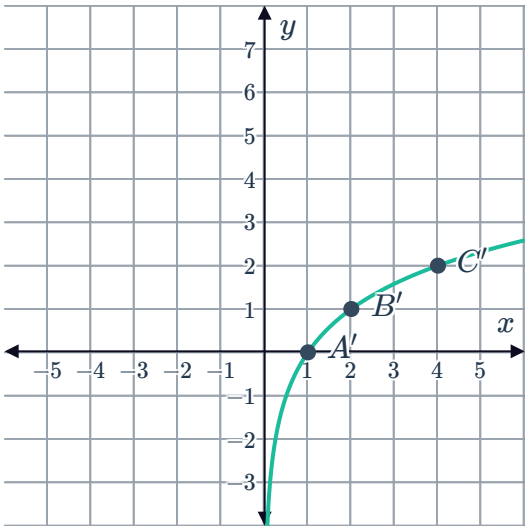
a

i $(0, 1)$ **ii** $(1, 2)$ **iii** $(2, 4)$

b

i $(1, 0)$ **ii** $(2, 1)$ **iii** $(4, 2)$

c



16

a $(-\infty, \infty)$ **b** $(0, \infty)$

c $(0, \infty)$ **d** $(-\infty, \infty)$

e The range of f^{-1} **f** The domain of f^{-1}

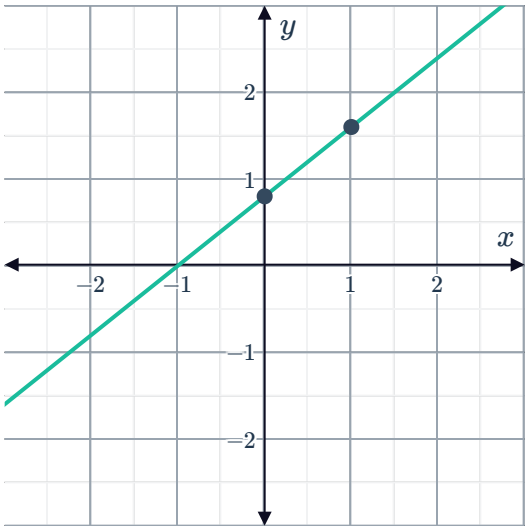
17

a $x \geq 2$ **b** $y \geq -5$ **c** $y = x^2 - 4x + 9$

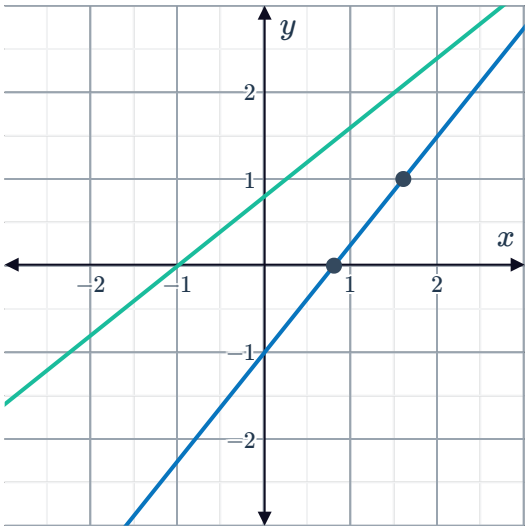
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18

a



b



c $(-\infty, \infty)$

d $(-\infty, \infty)$

e $(-\infty, \infty)$

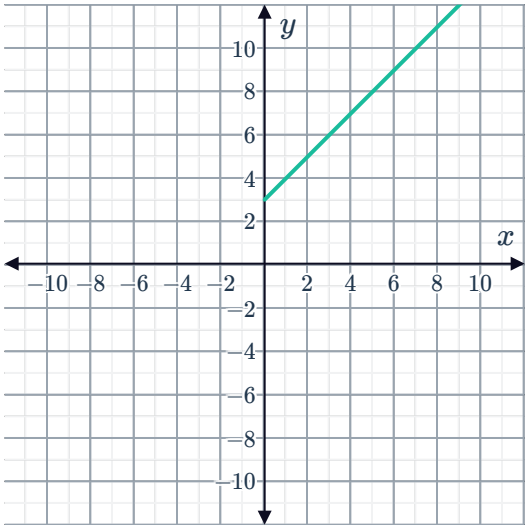
f $(-\infty, \infty)$

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19

a

i

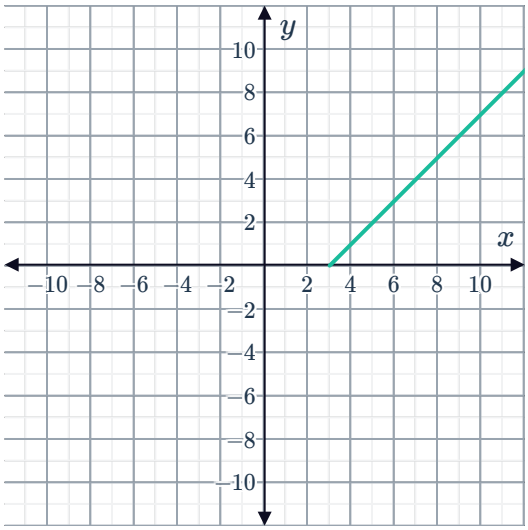


ii $f^{-1}(x) = x - 3$

iii $[3, \infty)$

iv $[0, \infty)$

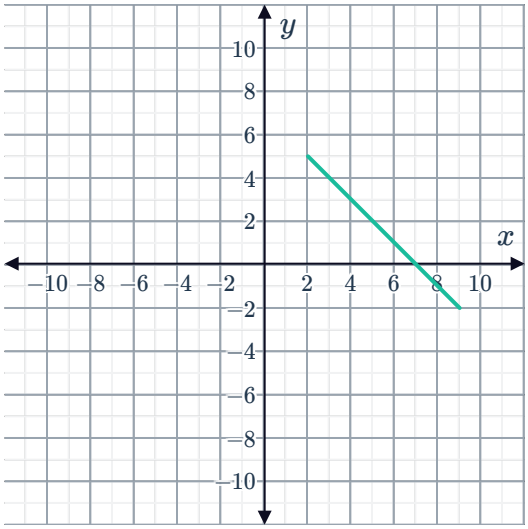
v



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b

i

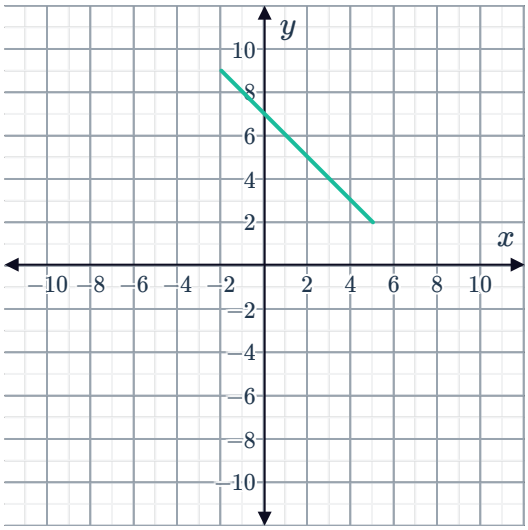


ii $f^{-1}(x) = 7 - x$

iii $[-2, 5]$

iv $[2, 9]$

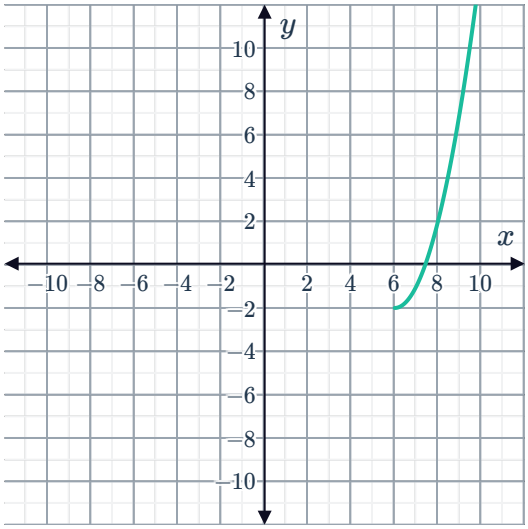
v



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c

i

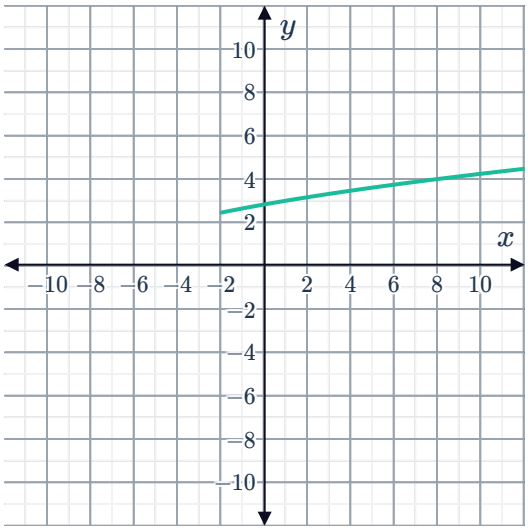


ii $f^{-1}(x) = \sqrt{x+2} + 6$

iii $[-2, \infty)$

iv $[6, \infty)$

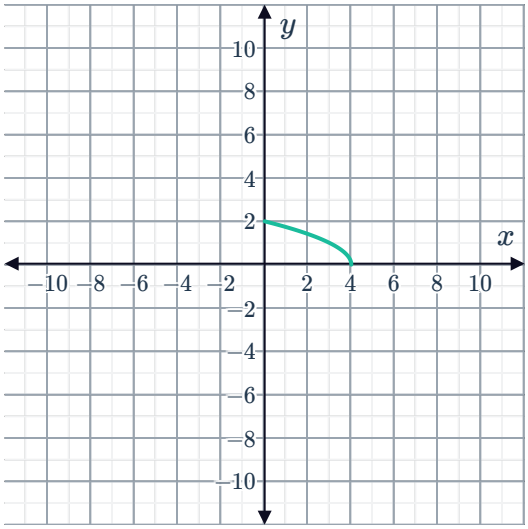
v



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d

i

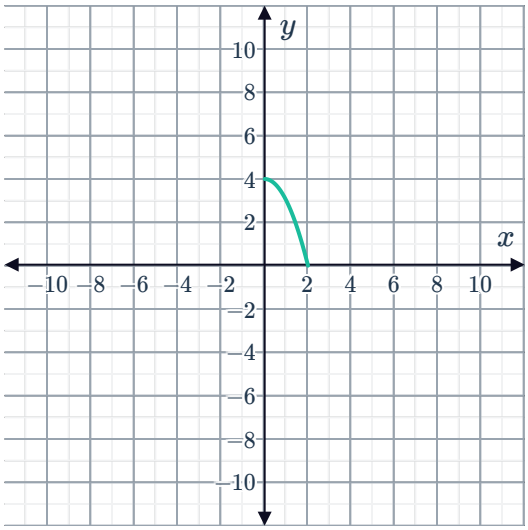


ii $f^{-1}(x) = 4 - x^2$

iii $(0, 2]$

iv $[0, 4)$

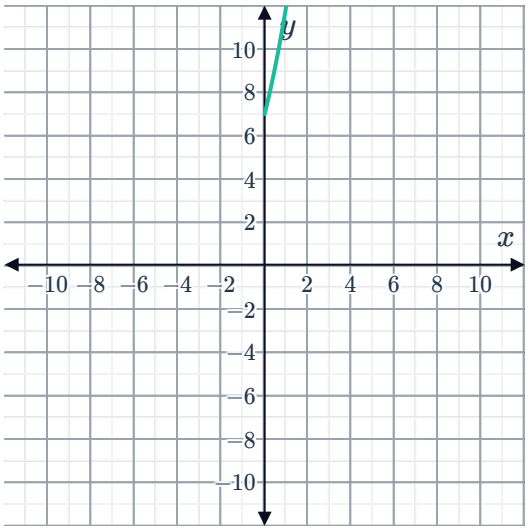
v



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e

i

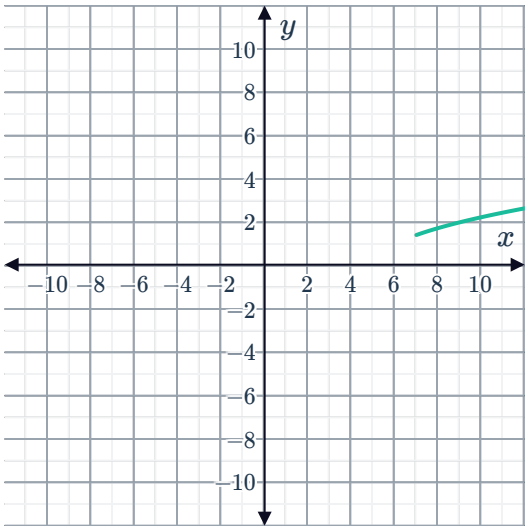


ii $f^{-1}(x) = \sqrt{x-3} - 2$

iii $[7, \infty)$

iv $[0, \infty)$

v



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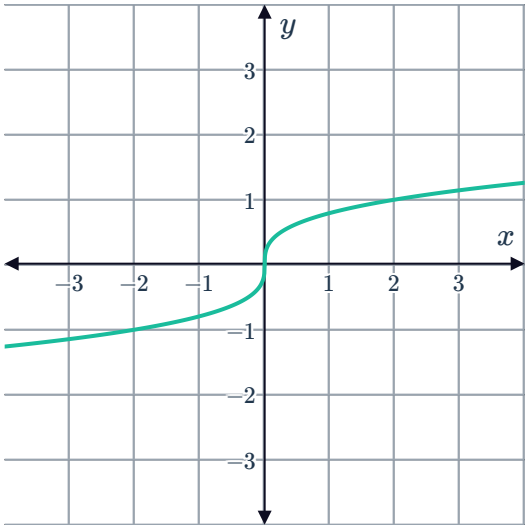
20

a Yes

b All real x

c All real y

d



21

a No

b All real x

c $x > 2$ and $x < -2$

d $y > 2$ or $y < -2$

22

a $T = 1.07A$

b $A = \dfrac{T}{1.07}$

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